



Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU

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FLOOD THREAT ADVISORY

To: **Brgy. Chairperson
Brgy. Concepcion Dos
City of Marikina
Metropolitan Manila**

Date: October 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Sapphire-Lilac	14° 39' 27.3" N, 121° 07' 13.3" E	Moderate	The police station is situated at the base of the gently sloping eastern hills of the barangay. Surface runoff from the highlands drain to this area as it drains further to Balante Creek, which acts as the boundary with Antipolo, Rizal. During Ondoy neck-high floodwaters were reported, while during Ulysses, around 4ft-high flood was estimated using the ruled post in the lower picture. Due to the improved drainage system and pumping stations within the barangay, waters subside after roughly 2 hours.
Rainbow St.	14° 39' 31.6" N, 121° 06' 56.1" E	High	Similar to the previous observation point, Rainbow St. also receives surface runoff from the "highlands" (elevated portion of barangay and neighboring barangays). During Ondoy, roughly 1m-high floodwaters were noted along the street. More recently (Ulysses), the flood height did not exceed that of the gutter.
Olive St.	14° 38' 16.6" N, 121° 07' 12.0" E	High	Various observations points within the periphery of Balante Creek are characterized by flood heights ranging from 1 to 1.5m, primarily caused by overwhelmed canals unable to drain to the creek, rarely due to overtopping of dike along the creek.
Green St.	14° 38' 23.8" N, 121° 06' 53.1" E	High	

Remarks / Recommendations:

- ✓ It is highly recommended to implement appropriate easements in construction of settlements or other infrastructure along rivers/creeks as specified in the Presidential Decree No. 1067 known as Water Code of the Philippines.
- ✓ Improvement and proper maintenance of existing drainage system in the barangay are recommended to prevent inundation of residential areas, critical facilities, and other assets. Avoid improper waste disposal, particularly along drainage canals as it may cause clogging and obstruction of water flow. Ensure regular clearing of natural and artificial drainage pathways, or prior to an impending tropical cyclone or other rain-bearing weather systems.

- ✓ Construction of appropriate and well-studied engineering measures (e.g., dike, flood wall, etc.) should be implemented to minimize the rate of erosion along rivers/creeks. Ensure that the structure follows engineering standards and has structural integrity. Also, damages in the structural mitigation measures should be reported to concerned agency/office responsible for the installation. The LGU is also encouraged to maintain such measures or reinforce identified damages, with the strong support of the barangay officials and active engagement of the communities and the private sector.
- ✓ Install a warning signage alerting motorists and residents of the danger of crossing bridges susceptible to flash flood during heavy rainfall events.
- ✓ The CDRRMO, together with the BDRRMO, should conduct Information, Education, and Communication (IEC) Campaign on flood hazards to flood-prone communities, particularly the households residing proximal to the river and irrigation canals.
- ✓ Strengthen and improve flood warning system with corresponding warning levels for pre-emptive evacuation purposes. The barangay should have constant communication with nearby and/or upland/upstream barangays during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas.
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of tropical cyclones and other prevailing weather disturbances. This may include City Disaster Risk Reduction Management Office (CDRRMO), Office of the Civil Defense (OCD) NCR, and other related agencies.

Prepared by:

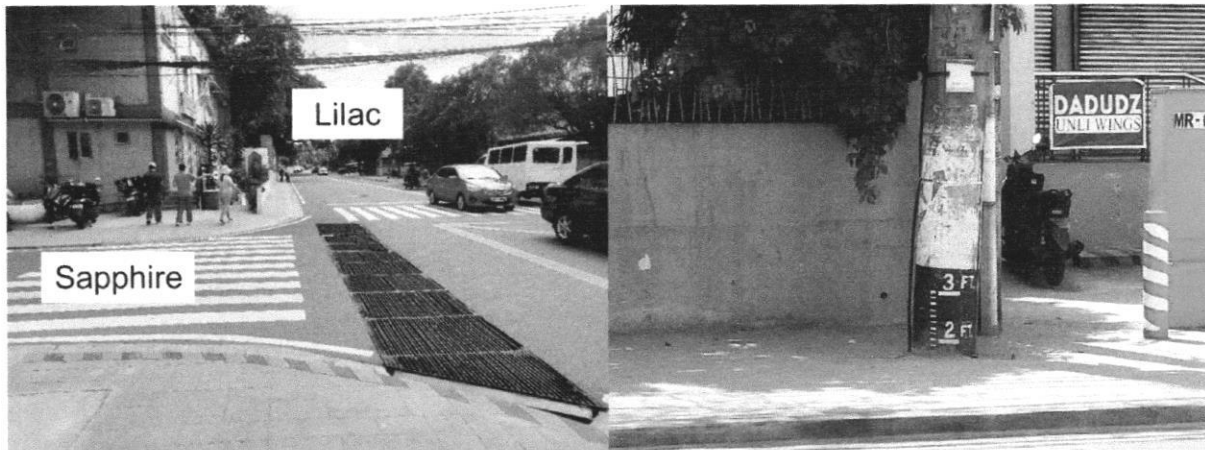

JULIUS VINCENT UMALI
Senior Geologist

MA. LOURDES STEPHANIE V. LEIDO
Senior Geologist


JETHRO B. CAPINO
Geologist II

Received by:
Designation:
Office:

BARANGAY CONCEPCION DOS



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Various observations points within the periphery of Balante Creek are characterized by flood heights ranging from 1 to 1.5m, primarily caused by overwhelmed canals unable to drain to the creek, rarely due to overtopping of dike along the creek (GPS Coordinates: 14° 38' 16.6 and 23.8" N, 121° 07' 06" and 12.0 and 53.1" E).

Recommendations:

- ✓ It is highly recommended to implement appropriate easements in construction of settlements or other infrastructure along rivers/creeks as specified in the Presidential Decree No. 1067 known as Water Code of the Philippines.
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